

Data Structure

Data structure is a method used by a computer to store and arrange data so that it can be accessed and changed in an efficient manner.

Examples:

- Array
- Linked List
- Stack
- Queue

Algorithm

An algorithm is a finite sequence of instructions that are used to solve a problem or accomplish a task.

Data Structure = Organizing data

Algorithm = The process

Linear Data Structure

Linear Data Structure: Data structure in which elements are placed one after another.

There are usually one or two elements preceding and/or following each element (other than the first and last).

Example: 10 -> 20 -> 30 -> 40 -> 50

Characteristics:

- The elements are listed in order.
- Data is typically accessible to traverse from one piece to another.
- More easily understood and more easily applied than other non-linear structures.

One-Dimensional Array

A one-dimensional array (1D array) is an array of elements of the same type that are stored sequentially. Each element is accessed using an index.

Example: `int[] numbers = { 10, 20, 30, 40, 50 };`

Common operations on an array:

- Accessing an element
- Updating an element
- Searching
- Sorting

Big O Notation

A way to measure the amount of work an algorithm requires as the size of the input grows is called big o notation.

It is used often to indicate the time complexity of an algorithm and occasionally to indicate the space complexity.

Basic Searching Algorithms

A searching algorithm is an algorithm used for the purpose of searching an ordered set of data for a certain element called the target.

Example:

Array: 10, 20, 30, 40, 50

Target: 40

There are two simple search algorithms:

- Linear Search
- Binary Search

Linear Search

Linear Search traverses' elements sequentially from the first to last to find the element.

Does not assume that the array is sorted.

Advantages

- Easy to comprehend.
- Easy to implement.
- Operates with unordered data.

Disadvantage

- May be slow for large arrays, as it could have to check each element.

Binary Search

In searching, a technique that halves the search area into half after every comparison many times is used is called Binary Search. Note that data must be sorted.

Advantages

- Linear search is more time consuming than Binary search for large, sorted arrays.
- Halves the area to be searched after each comparison.

Disadvantage

- The data should be sorted.

Citations/References

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